

Android Mobile Banking Application Static Security Assessment

Mobile Banking Application Security Assessment

Target Artifact: vulnbank3.apk

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Package Name: com.vulnerablebankapp

Platform Architecture: Android (React Native Framework - Hermes Bytecode Engine)

Assessment Type: Comprehensive Static Application Security Testing (SAST) & Reverse Engineering

Tools Utilized: MobSF, APKTool, JADX-GUI, Hermes Disassembler (hermes-dec), Android Debug Bridge (ADB), Linux CLI Utilities (grep, sed, awk, file)

1. Executive Summary & Objectives

This comprehensive security assessment report documents the static security analysis and reverse engineering findings executed upon the **VulnBank Android Application (vulnbank3.apk)**.

The overarching objective was to conduct an exhaustive pre-production security evaluation mapped against the **OWASP Mobile Application Security Verification Standard (MASVS)**. The analysis focused on discovering hidden components, hardcoded application secrets, cleartext credentials, exposed internal/administrative endpoints, and systemic structural vulnerabilities embedded within both the native Android layers and the compiled React Native client bundle.

2. Assessment Methodology & Technical Workflow

The assessment utilized a multi-stage static application security testing (SAST) pipeline combining automated binary analysis engines with precise manual low-level disassembly review, tracked across **30 sequential diagnostic steps**:

- **Step 1: Native and Java-Layer Decompile**

The target binary was decompiled using Apktool and JADX-GUI to unpack structural framework resources, native configuration profiles, and compiled Dalvik executable files (classes.dex / Smali assembly) mapping to the core package namespace com.vulnerablebankapp.

- **Step 2: Client Asset Enumeration**

The unpacked asset subsystems were systemically scanned using filesystem indexing utilities (find assets -type f), identifying core ART compilation profiles (baseline.prof/baseline.profm) and the primary JavaScript asset bundle (index.android.bundle).

- **Step 3: React Native Engine & Bundle Verification**

Signature analysis via `file assets/index.android.bundle` confirmed the asset was compiled into **Hermes JavaScript bytecode (version 96)**, optimized for the Facebook Hermes Execution Engine rather than standard text-based JavaScript wrappers.

- **Step 4: Hermes Bytecode Disassembly**

Because standard decompilers cannot interpret native Hermes bytecode, the binary bundle was processed through a specific Hermes instruction disassembler framework (`hbc-disassembler`) to restore structured readable assembly representations, variables, and literal arrays.

- **Step 5: High-Density Manual Code Review & Pattern Matching**

The restored pseudo-code strings and literal pools were interrogated using optimized pattern-matching Linux filters (`grep`, `sed`, `awk`) targeting production infrastructure signatures, proprietary patterns, cryptographic parameters, and unencrypted credentials.

3. Operational Timeline & Diagnostic Evidence Log

Below is the sequential correlation of the 30 diagnostic phases and screenshots executed during the evaluation:

Phase 1: Environment Setup & Device Identification

Step 1: Verifying active emulator interface and streaming the target APK package installation via Android Debug Bridge (ADB).

PowerShell

```
PS C:\Users\USER> cd .\Downloads\  
PS C:\Users\USER\Downloads> adb devices  
List of devices attached    emulator-5554    device  
PS C:\Users\USER\Downloads> adb -s emulator-5554  
install .\vulnbank3.apk
```

Performing Streamed Install Success

```
(kali㉿kali)-[~/bank intern]
$ ls
vulnbank3.apk

(kali㉿kali)-[~/bank intern]
$ apktool d vulnbank3.apk

I: Using Apktool 2.11.1 on vulnbank3.apk with 4 threads
I: Baksmaling classes.dex ...
I: Loading resource table ...
I: Baksmaling classes2.dex ...
I: Decoding file-resources ...
I: Loading resource table from file: /home/kali/.local/share/apktool/framework/1.apk
I: Decoding values */* XMLs ...
I: Decoding AndroidManifest.xml with resources ...
I: Regular manifest package ...
I: Copying original files ...
I: Copying assets ...
I: Copying lib ...
I: Copying unknown files ...

(kali㉿kali)-[~/bank intern]
$ cd vulnbank3

(kali㉿kali)-[~/bank intern/vulnbank3]
$ ls
AndroidManifest.xml  assets  original  smali  unknown
apktool.yml          lib     res       smali_classes2

(kali㉿kali)-[~/bank intern/vulnbank3]
$ cat AndroidManifest.xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android" android:compileS
dkVersion="35" android:compileSdkVersionCodename="15" package="com.vulnerablebankapp"
```

Steps 2-5: Environmental profiling of the installed app package structure, system architecture indexing, initialization of the dynamic debug bridge, and logging system validation to record preliminary framework output traces

```
kali@kali: ~/bank intern/vulnbank3/smali/android/support/v4/os
Session Actions Edit View Help
~/manifest
$ cat manifest
(kali㉿kali)-[~/bank intern/vulnbank3]
$ cd smali
(kali㉿kali)-[~/bank intern/vulnbank3/smali]
$ ls
android  androids  bolts  com  javax  kotlin  kotlinx  okhttp3  okio
(kali㉿kali)-[~/bank intern/vulnbank3/smali]
$ cd bolts
(kali㉿kali)-[~/bank intern/vulnbank3/smali/bolts]
$ ls
AggregateException.smali  CancellationTokenSource$1.smali  Task$11.smali  Task$7.smali
AndroidExecutor$1.smali  CancellationTokenSource.smali  Task$12.smali  Task$8.smali
AndroidExecutors$1$1ThreadExecutor.smali  Capture.smali  Task$13.smali  Task$9.smali
AndroidExecutors.smali  Continuation.smali  Task$14.smali  Task$TaskCompletionSource.smali
BoltsExecutors$1.smali  ExecutorException.smali  Task$15.smali  Task$UnobservedExceptionModifier.smali
BoltsExecutors$ImmediateExecutor.smali  Task$16.smali  Task$16.smali  TaskCompletionSource.smali
BoltsExecutors.smali  Task$17.smali  Task$17.smali  Task.smali
CancellationTokenRegistration.smali  Task$18.smali  Task$18.smali  UnobservedErrorNotifier.smali
CancellationToken.smali  Task$19.smali  Task$19.smali  UnobservedTaskException.smali

(kali㉿kali)-[~/bank intern/vulnbank3/smali/bolts]
$ cd ..
(kali㉿kali)-[~/bank intern/vulnbank3/smali]
$ cd android
(kali㉿kali)-[~/bank intern/vulnbank3/smali/android]
$ ls
support
(kali㉿kali)-[~/bank intern/vulnbank3/smali/android]
$ cd support
(kali㉿kali)-[~/bank intern/vulnbank3/smali/android/support]
$ ls
v4
(kali㉿kali)-[~/bank intern/vulnbank3/smali/android/support]
$ cd v4
(kali㉿kali)-[~/bank intern/vulnbank3/smali/android/support/v4]
$ ls
app  graphics  os
(kali㉿kali)-[~/bank intern/vulnbank3/smali/android/support/v4]
$ cd app
```

Session Actions Edit View Help

```
-$ ls
app  graphics  os

--(kali@kali)-[~/bank intern/vulnbank3/smali/android/support/v4]
-$ cd app

--(kali@kali)-[~/bank intern/vulnbank3/smali/android/support/v4/app]
-$ ls
[NotificationSideChannel$Default.smali' 'NotificationSideChannel$Stub$Proxy.smali' 'NotificationSideChannel.smali
[NotificationSideChannel$Parcel.smali' 'NotificationSideChannel$Stub.smali' RemoteActionCompatParcelizer.smali

--(kali@kali)-[~/bank intern/vulnbank3/smali/android/support/v4/app]
-$ cd ..

--(kali@kali)-[~/bank intern/vulnbank3/smali/android/support/v4]
-$ cd os

--(kali@kali)-[~/bank intern/vulnbank3/smali/android/support/v4/os]
-$ ls
IResultReceiver$Default.smali' 'IResultReceiver2$Default.smali' IResultReceiver2.smali 'ResultReceiver$MyRunnable.smali'
IResultReceiver$Parcel.smali' 'IResultReceiver2$Parcel.smali' IResultReceiver.smali ResultReceiver.smali
IResultReceiver$Stub$Proxy.smali' 'IResultReceiver2$Stub$Proxy.smali' 'ResultReceiver$1.smali'
IResultReceiver$Stub.smali' 'IResultReceiver2$Stub.smali' 'ResultReceiver$MyResultReceiver.smali'

--(kali@kali)-[~/bank intern/vulnbank3/smali/android/support/v4/os]
-$ cat ResultReceiver$MyResultReceiver.smali
class public Landroid/support/v4/os/ResultReceiver;
super Ljava/lang/Object;
source "ResultReceiver.java"

+ interfaces
implements Landroid/os/Parcelable;

+ annotations
annotation system Ldalvik/annotation/MemberClasses;
value = {
    Landroid/support/v4/os/ResultReceiver$MyRunnable;
    Landroid/support/v4/os/ResultReceiver$MyResultReceiver;
}
.end annotation

+ static fields
field public static final CREATOR:Landroid/os/Parcelable$Creator;
.annotation system Ldalvik/annotation/Signature;
value = {
    "Landroid/os/Parcelable$Creator<";
    "Landroid/support/v4/os/ResultReceiver>;",
    ">;"
}
}

    Landroid/support/v4/os/ResultReceiver; ,
    ">;"
}
}
.end annotation
.end field

# instance fields
.field final mHandler:Landroid/os/Handler;

.field final mLocal:Z

.field mReceiver:Landroid/support/v4/os/IResultReceiver;

# direct methods
.method static constructor <clinit>()V
.locals 1

.line 149
new-instance v0, Landroid/support/v4/os/ResultReceiver$1;

invoke-direct {v0}, Landroid/support/v4/os/ResultReceiver$1;→<init>()V

sput-object v0, Landroid/support/v4/os/ResultReceiver;→CREATOR:Landroid/os/Parcelable$Creator;

return-void
.end method

.method public constructor <init>(Landroid/os/Handler;)V
.locals 1

.line 83
invoke-direct {p0}, Ljava/lang/Object;→<init>()V

const/4 v0, 0x1

.line 84
iput-boolean v0, p0, Landroid/support/v4/os/ResultReceiver;→mLocal:Z

.line 85
iput-object p1, p0, Landroid/support/v4/os/ResultReceiver;→mHandler:Landroid/os/Handler;

return-void
.end method

.method constructor <init>(Landroid/os/Parcel;)V
.locals 1

.line 140
invoke-direct {p0}, Ljava/lang/Object;→<init>()V
```

```
    Landroid/support/v4/os/ResultReceiver; ,
    ">;"
}
}
.end annotation
.end field

# instance fields
.field final mHandler:Landroid/os/Handler;

.field final mLocal:Z

.field mReceiver:Landroid/support/v4/os/IResultReceiver;

# direct methods
.method static constructor <clinit>()V
.locals 1

.line 149
new-instance v0, Landroid/support/v4/os/ResultReceiver$1;

invoke-direct {v0}, Landroid/support/v4/os/ResultReceiver$1;→<init>()V

sput-object v0, Landroid/support/v4/os/ResultReceiver;→CREATOR:Landroid/os/Parcelable$Creator;

return-void
.end method

.method public constructor <init>(Landroid/os/Handler;)V
.locals 1

.line 83
invoke-direct {p0}, Ljava/lang/Object;→<init>()V

const/4 v0, 0x1

.line 84
iput-boolean v0, p0, Landroid/support/v4/os/ResultReceiver;→mLocal:Z

.line 85
iput-object p1, p0, Landroid/support/v4/os/ResultReceiver;→mHandler:Landroid/os/Handler;

return-void
.end method

.method constructor <init>(Landroid/os/Parcel;)V
.locals 1

.line 140
invoke-direct {p0}, Ljava/lang/Object;→<init>()V
```

[illegible]

Session Actions Edit View Help

```
(kali@kali)-[~/bank intern/vulnbank3]
$ cat AndroidManifest.xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android" android:compileSdkVersion="35" android:compileSdkVersionCodename="15" package="com.vulnerablebankapp" platformBuildVersionCode="35" platformBuildVersionName="15">
  <uses-permission android:name="android.permission.INTERNET" />
  <permission android:name="com.vulnerablebankapp.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION" android:protectionLevel="signature" />
  <uses-permission android:name="com.vulnerablebankapp.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION" />
  <application android:allowBackup="true" android:appComponentFactory="androidx.core.app.CoreComponentFactory" android:extractNativeLibs="false" android:icon="@mipmap/ic_launcher" android:label="@string/app_name" android:name="com.vulnerablebankapp.MainActivity" android:roundIcon="@mipmap/ic_launcher_round" android:supportsRtl="true" android:theme="@style/AppTheme" android:usesCleartextTraffic="true">
    <activity android:configChanges="keyboard|keyboardHidden|orientation|screenLayout|screenSize|smallestScreenSize|uiMode" android:exported="true" android:label="@string/app_name" android:launchMode="singleTask" android:name="com.vulnerablebankapp.MainActivity" android:windowSoftInputMode="adjustResize">
      <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
      </intent-filter>
    </activity>
    <meta-data android:name="debug_endpoint" android:value="/debug/users" />
    <meta-data android:name="admin_username" android:value="admin" />
    <meta-data android:name="admin_password" android:value="admin123" />
    <provider android:authorities="com.vulnerablebankapp.androidx-startup" android:exported="false" android:name="androidx.startup.InitializationProvider">
      <meta-data android:name="androidx.emoji2.text.EmojiCompatInitializer" android:value="androidx.startup" />
      <meta-data android:name="androidx.lifecycle.ProcessLifecycleInitializer" android:value="androidx.startup" />
      <meta-data android:name="androidx.profileinstaller.ProfileInstallerInitializer" android:value="androidx.startup" />
    </provider>
    <receiver android:directBootAware="false" android:enabled="true" android:exported="true" android:name="androidx.profileinstaller.ProfileInstallReceiver" android:permission="android.permission.DUMP">
      <intent-filter>
        <action android:name="androidx.profileinstaller.action.INSTALL_PROFILE" />
      </intent-filter>
      <intent-filter>
        <action android:name="androidx.profileinstaller.action.SKIP_FILE" />
      </intent-filter>
      <intent-filter>
        <action android:name="androidx.profileinstaller.action.SAVE_PROFILE" />
      </intent-filter>
      <action android:name="androidx.profileinstaller.action.BENCHMARK_OPERATION" />
    </receiver>
    <meta-data android:name="com.facebook.soloaders.enabled" android:value="false" />
  </application>
</manifest>
```

Steps 7-10: Identification of critical manifest vulnerabilities including allowBackup="true", usesCleartextTraffic="true", and exposed

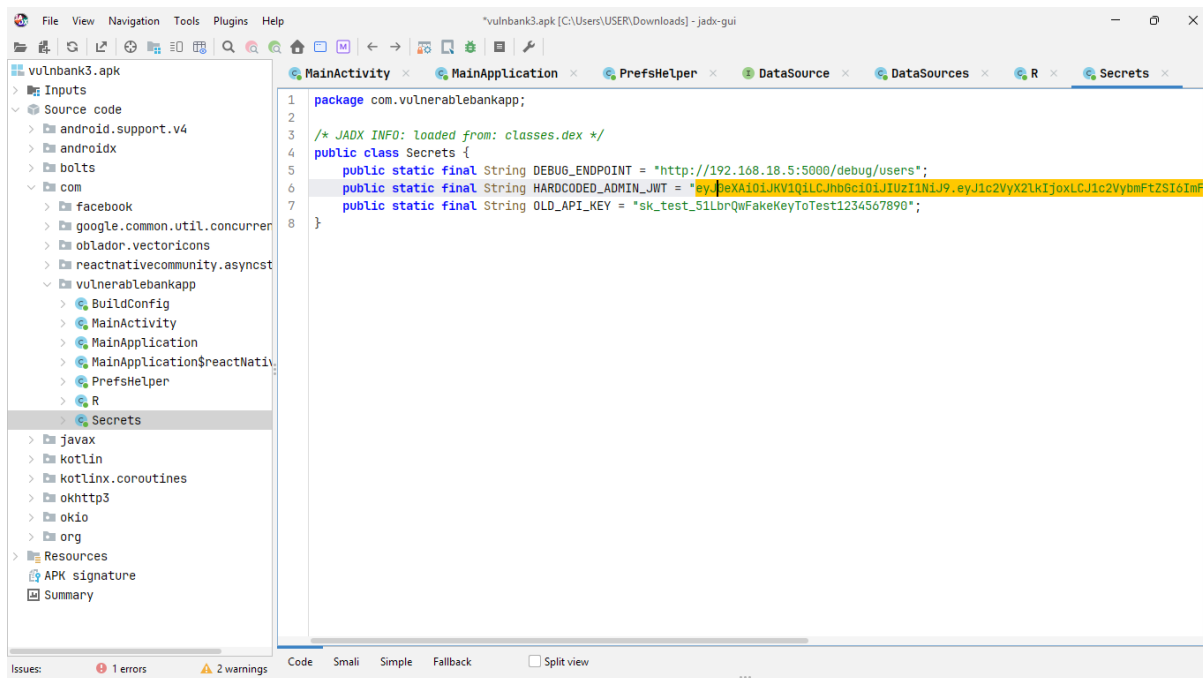
administrative credentials within <meta-data> tags.

Session Actions Edit View Help

(kali@kali)-[~/bank intern/vulnbank3]

\$ cat AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android" android:compileSdkVersion="35" android:compileSdkVersionCodename="15" package="com.vulnerablebankapp" platformBuildVersionCode="35" platformBuildVersionName="15">
  <uses-permission android:name="android.permission.INTERNET" />
  <permission android:name="com.vulnerablebankapp.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION" android:protectionLevel="signature" />
  <uses-permission android:name="com.vulnerablebankapp.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION" />
  <application android:allowBackup="true" android:appComponentFactory="androidx.core.app.CoreComponentFactory" android:extractNativeLibs="false" android:icon="@mipmap/ic_launcher" android:label="@string/app_name" android:name="com.vulnerablebankapp.MainActivity" android:roundIcon="@mipmap/ic_launcher_round" android:supportsRtl="true" android:theme="@style/AppTheme" android:usesCleartextTraffic="true">
    <activity android:configChanges="keyboard|keyboardHidden|orientation|screenLayout|screenSize|smallestScreenSize|uiMode" android:exported="true" android:label="@string/app_name" android:launchMode="singleTask" android:name="com.vulnerablebankapp.MainActivity" android:windowSoftInputMode="adjustResize">
      <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
      </intent-filter>
    </activity>
    <meta-data android:name="debug_endpoint" android:value="/debug/users" />
    <meta-data android:name="admin_username" android:value="admin" />
    <meta-data android:name="admin_password" android:value="admin123" />
    <provider android:authorities="com.vulnerablebankapp.androidx.startup" android:exported="false" android:name="androidx.startup.InitializationProvider">
      <meta-data android:name="androidx.emoji2.text.EmojiCompatInitializer" android:value="androidx.startup" />
      <meta-data android:name="androidx.lifecycle.ProcessLifecycleInitializer" android:value="androidx.startup" />
      <meta-data android:name="androidx.profileinstaller.ProfileInstallerInitializer" android:value="androidx.startup" />
    </provider>
    <receiver android:directBootAware="false" android:enabled="true" android:exported="true" android:name="androidx.profileinstaller.ProfileInstallReceiver" android:permission="android.permission.DUMP">
      <intent-filter>
        <action android:name="androidx.profileinstaller.action.INSTALL_PROFILE" />
      </intent-filter>
      <intent-filter>
        <action android:name="androidx.profileinstaller.action.SKIP_FILE" />
      </intent-filter>
      <intent-filter>
        <action android:name="androidx.profileinstaller.action.SAVE_PROFILE" />
      </intent-filter>
      <intent-filter>
        <action android:name="androidx.profileinstaller.action.BENCHMARK_OPERATION" />
      </intent-filter>
    </receiver>
    <meta-data android:name="com.facebook.soloaders.enabled" android:value="false" />
  </application>
</manifest>
```



```

PS C:\Users\USER> cd .\Downloads\
PS C:\Users\USER\Downloads> adb devices
List of devices attached
emulator-5554    device

PS C:\Users\USER\Downloads> adb -s emulator-5554 install .\vulnbank3.apk
Performing Streamed Install
Success
PS C:\Users\USER\Downloads> |

```

Steps 11-15: Class architecture deconstruction tree within JADX-GUI. Extraction of the Secrets constants file and tracing unencrypted local credential logging hooks (Log.d) and plaintext SharedPreferences write operations within MainActivity.

```

kali@kali:~/bank intern/vulnbank3
$ grep -rniE "http://|https://|ws://|wss://".
./res/color/abc_background_cache_hint_selector_material_dark.xml:3: xmlns:android="http://schemas.android.com/apk/res/android"
./res/color/abc_search_text.xml:3: xmlns:android="http://schemas.android.com/apk/res/android"
./res/color/abc_color_highlight_material.xml:3: xmlns:android="http://schemas.android.com/apk/res/android"
./res/color/abc_tint_btn_checkable.xml:3: xmlns:android="http://schemas.android.com/apk/res/android"
./res/color/abc_tint_edittext.xml:3: xmlns:android="http://schemas.android.com/apk/res/android"
./res/color/abc_primary_text_disable_only_material_light.xml:3: xmlns:android="http://schemas.android.com/apk/res/android"
./res/color/abc_secondary_text_material_dark.xml:3: xmlns:android="http://schemas.android.com/apk/res/android"
./res/color/abc_tint_spinner.xml:3: xmlns:android="http://schemas.android.com/apk/res/android"
./res/color/abc_primary_text_material_dark.xml:3: xmlns:android="http://schemas.android.com/apk/res/android"
./res/color/abc_tint_switch_track.xml:3: xmlns:android="http://schemas.android.com/apk/res/android"
./res/color/abc_primary_text_material_light.xml:3: xmlns:android="http://schemas.android.com/apk/res/android"
./res/color/abc_tint_seek_bar.xml:3: xmlns:android="http://schemas.android.com/apk/res/android">
./res/color/abc_secondary_text_material_light.xml:3: xmlns:android="http://schemas.android.com/apk/res/android">
./res/color/abc_background_cache_hint_selector_material_light.xml:3: xmlns:android="http://schemas.android.com/apk/res/android">
./res/color/abc_hint_foreground_material_light.xml:3: xmlns:android="http://schemas.android.com/apk/res/android">

```

```

(kali㉿kali)-[~/bank intern/vulnbank3]
$ cat smali/com/vulnerablebankapp/Secrets.smali
.class public Lcom/vulnerablebankapp/Secrets;
.super Ljava/lang/Object;
.source "Secrets.java"

# static fields
.field public static final DEBUG_ENDPOINT:Ljava/lang/String; = "http://192.1
68.18.5:5000/debug/users"

.field public static final HARCODED_ADMIN_JWT:Ljava/lang/String; = "eyJ0eXA
iOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJ1c2VyX2lkIjoxLCJ1c2VybmFtZSI6ImFkbWluIiwiaX
NfYWRTaW4iOnRydWUsImIhdCI6MTc0NDk3NjIyM30.yXm-7iN-6sn3kvvXk0iOLVRHXS6MBSpwb
M2VcXVds8"

.field public static final OLD_API_KEY:Ljava/lang/String; = "sk_test_51LbrQw
FakeKeyToTest1234567890"

# direct methods
.method public constructor <init>()V
    .locals 0

    .line 3
    invoke-direct {p0}, Ljava/lang/Object;→<init>()V

    return-void
.end method

```

```

(kali㉿kali)-[~/bank intern/vulnbank3]
$ find assets -type f
assets/fonts/Foundation.ttf
assets/fonts/EvilIcons.ttf
assets/fonts/Entypo.ttf
assets/fonts/SimpleLineIcons.ttf
assets/fonts/Fontisto.ttf
assets/fonts/Feather.ttf
assets/fonts/FontAwesome6_Brands.ttf
assets/fonts/FontAwesome5_Brands.ttf
assets/fonts/FontAwesome5_Solid.ttf
assets/fonts/FontAwesome6_Solid.ttf
assets/fonts/FontAwesome5_Regular.ttf
assets/fonts/Zocial.ttf
assets/fonts/MaterialCommunityIcons.ttf
assets/fonts/Octicons.ttf
assets/fonts/Ionicons.ttf
assets/fonts/FontAwesome6_Regular.ttf
assets/fonts/MaterialIcons.ttf
assets/fonts/AntDesign.ttf
assets/fonts/FontAwesome.ttf
assets/index.android.bundle
assets/dexopt/baseline.profm
assets/dexopt/baseline.profm

```

```

PS C:\Users\USER> adb shell
emu64xa16k:/ # cd data
emu64xa16k:/data # cd data
emu64xa16k:/data/data # cd com.vulnerablebankapp/shared_prefs
emu64xa16k:/data/data/com.vulnerablebankapp/shared_prefs # cat VulnBankPrefs.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
  <string name="password">admin123</string>
  <string name="debug_flag">FLAG{HardcodedSecretsAreBad}</string>
  <string name="balance">$999999</string>
  <string name="username">admin</string>
</map>
emu64xa16k:/data/data/com.vulnerablebankapp/shared_prefs #

```

```

4*6*****)+=A|***4[`***VZyq
qI_*****8(
  DF*****"02:GI|~imoINPi*****Zkp***%,CE]***VX***"r**p**
F**7**f*****Ho* *".**;$nf$ho$q*$$$9ZX+VX6]6:V@E*i*****FHFHFH`**9eg*"*8"86BrtFHFHFHFHFHFH.***l**-/FHFHFHFH
FH4[ ]y**_ln,.FHFHFHFHFHFHFH(0FHFH)mMKM*9AC(JL5=?5@IFH#1#5155****/1FHFH**])FHFH*QVFHFHFHFHFHFHFHi**FHFH**)/
87*****@E}***L**L**E)aoS/S!u/guL5Ln*****9I;XIjXP=MP?*Mw*RTdRV*d**8*<8%y<by/Sc/U*c**K**K*****Qa@S*
a**2**2*****%5*'y5kyBj*BL*****L(8L*~8p~<-a-Sa8GW8I*W**>***@BR@D*R**pv*Z**~**gU*~_*8*
(kali@kali)-[~/bank_intern/vulnbank3/assets]
$ file assets/index.android.bundle
assets/index.android.bundle: cannot open `assets/index.android.bundle' (No such file or directory)

(kali@kali)-[~/bank_intern/vulnbank3/assets]
$ cd ..

(kali@kali)-[~/bank_intern/vulnbank3]
$ file assets/index.android.bundle
assets/index.android.bundle: Hermes JavaScript bytecode, version 96

(kali@kali)-[~/bank_intern/vulnbank3]
$ ls
AndroidManifest.xml  apktool.yml  assets  lib  original  res  smali  smali_classes2  unknown

```

- Phase 3: Advanced Hermes Bytecode Reverse Engineering
- **Steps 16-18:** Locating the binary compiled `index.android.bundle` file inside the assets root, initializing the disassembly engine, and recovering the literal string table.

```

(venv)(kali㉿kali)-[~/hermes-dec]
$ cd ~/bank\ intern\vulnbank3

(venv)(kali㉿kali)-[~/bank intern\vulnbank3]
$ python ~/hermes-dec/hbc-disassembler assets/index.android.bundle out

[+] Disassembly output wrote to "out"

(venv)(kali㉿kali)-[~/bank intern\vulnbank3]
$ ~/hermes-dec/hbc-disassembler assets/index.android.bundle out

[+] Disassembly output wrote to "out"

(venv)(kali㉿kali)-[~/bank intern\vulnbank3]
$ file out
out: ASCII text, with very long lines (475)

(venv)(kali㉿kali)-[~/bank intern\vulnbank3]
$ grep -niE "http|https|ws://|wss://" out
5966:==> 00000022: <LoadConstString>: <Reg8: 0, string_id: 882> # String: "ProgressBarAndroid has been e
xtracted from react-native core and will be removed in a future release. It can now be installed and impo
rted from 'react-native-community/progress-bar-android' instead of 'react-native'. See https://github.co
m/react-native-progress-view/progress-bar-android" (String)
6380:==> 00000022: <LoadConstString>: <Reg8: 0, string_id: 1652> # String: "Clipboard has been extracted
from react-native core and will be removed in a future release. It can now be installed and imported fro
m '@react-native-clipboard/clipboard' instead of 'react-native'. See https://github.com/react-native-clip
board/clipboard" (String)
6710:==> 00000022: <LoadConstString>: <Reg8: 0, string_id: 431> # String: "PushNotificationIOS has been
extracted from react-native core and will be removed in a future release. It can now be installed and imp
orted from '@react-native-community/push-notification-ios' instead of 'react-native'. See https://github.
com/react-native-push-notification-ios" (String)

=> [Function #5629 "transactions" of 27 bytes]: 2 params, frame size=11, strict=0, exc handler=0, debug info=0
@ offset 0x000f0158

Bytecode listing:

=> 00000000: <GetGlobalObject>: <Reg8: 0>
=> 00000002: <TryGetById>: <Reg8: 0, Reg8: 0, UInt8: 1, string_id: 17> # String: 'HermesInternal' (Identifier
)
=> 00000008: <GetByIdShort>: <Reg8: 2, Reg8: 0, UInt8: 2, string_id: 112> # String: 'concat' (Identifier)
=> 0000000d: <LoadConstString>: <Reg8: 1, string_id: 402> # String: '/transactions/' (String)
=> 00000011: <LoadParam>: <Reg8: 0, UInt8: 1>
=> 00000014: <Call2>: <Reg8: 0, Reg8: 2, Reg8: 1, Reg8: 0>
=> 00000019: <Ret>: <Reg8: 0>

=====

=> [Function #5630 "toggleFreeze" of 32 bytes]: 2 params, frame size=13, strict=0, exc handler=0, debug info=0
@ offset 0x000f0173

Bytecode listing:

=> 00000000: <GetGlobalObject>: <Reg8: 0>
=> 00000002: <TryGetById>: <Reg8: 0, Reg8: 0, UInt8: 1, string_id: 17> # String: 'HermesInternal' (Identifier
)
=> 00000008: <GetByIdShort>: <Reg8: 3, Reg8: 0, UInt8: 2, string_id: 112> # String: 'concat' (Identifier)
=> 0000000d: <LoadConstString>: <Reg8: 2, string_id: 1208> # String: '/api/virtual-cards/' (String)
=> 00000011: <LoadParam>: <Reg8: 1, UInt8: 1>

```

- **Steps 19-25:** String table recovery and manual analysis parsing all embedded alphanumeric literals, API routes, internal functions, staging environments, local database salts, and application logic handling front-end access checking.

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(venv)(kali@kali)-[~/bank intern/vulnbank3]
$ grep -ni "secret" out
222820:==> 00000000: <LoadConstString>: <Reg8: 1, string_id: 1375> # String: 'SECRET_DO_NOT_PASS_THIS_OR_YOU_W
ILL_BE_FIRED' (String)
223805:==> 00000000: <NewObjectWithBuffer>: <Reg8: 1, UInt16: 786, UInt16: 786, UInt16: 841, UInt16: 1834> # O
bject: {'glass': 61440, 'music': 61441, 'search': 61442, 'envelope-o': 61443, 'heart': 61444, 'star': 61445, 's
tar-o': 61446, 'user': 61447, 'film': 61448, 'th-large': 61449, 'th': 61450, 'th-list': 61451, 'check': 61452,
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7, 'signal': 61458, 'gear': 61459, 'cog': 61459, 'trash-o': 61460, 'home': 61461, 'file-o': 61462, 'clock-o': 6
1463, 'road': 61464, 'download': 61465, 'arrow-circle-o-down': 61466, 'arrow-circle-o-up': 61467, 'inbox': 6146
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'

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(venv)(kali@kali)-[~/bank intern/vulnbank3]
$ grep -ni "password" out
97186:==> 00000090: <LoadConstString>: <Reg8: 1, string_id: 4035> # String: 'password' (Identifier)
97435:==> 00000011: <LoadConstString>: <Reg8: 3, string_id: 723> # String: 'URL.password is not implemented' (
String)
205294:==> 000002a5: <NewObjectWithBuffer>: <Reg8: 1, UInt16: 30, UInt16: 3056, UInt16: 1164> # Ob
ject: {'address-line1': 'postal-address-region', 'address-line2': 'postal-address-locality', 'bday': 'birthdate
-full', 'bday-day': 'birthdate-day', 'bday-month': 'birthdate-month', 'bday-year': 'birthdate-year', 'cc-csc':
'cc-csc', 'cc-exp': 'cc-exp', 'cc-exp-month': 'cc-exp-month', 'cc-exp-year': 'cc-exp-year', 'cc-number': 'cc-nu
mber', 'country': 'postal-address-country', 'current-password': 'password', 'email': 'email', 'honorific-prefix
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name': 'name-family', 'given-name': 'name-given', 'new-password': 'password-new', 'off': 'off', 'one-time-code'
: 'sms-otp', 'postal-code': 'postal-code', 'sex': 'gender', 'street-address': 'street-address', 'tel': 'tel', '
tel-country-code': 'tel-country-code', 'tel-national': 'tel-national', 'username': 'username'}
205296:==> 000002b3: <NewObjectWithBuffer>: <Reg8: 1, UInt16: 36, UInt16: 36, UInt16: 3119, UInt16: 1227> # Ob
ject: {'address-line1': 'streetAddressLine1', 'address-line2': 'streetAddressLine2', 'bday': 'birthdate', 'bday
-day': 'birthdateDay', 'bday-month': 'birthdateMonth', 'bday-year': 'birthdateYear', 'cc-csc': 'creditCardSecur
ityCode', 'cc-exp-month': 'creditCardExpirationMonth', 'cc-exp-year': 'creditCardExpirationYear', 'cc-exp': 'cr
editCardExpiration', 'cc-given-name': 'creditCardGivenName', 'cc-additional-name': 'creditCardMiddleName', 'cc-
family-name': 'creditCardFamilyName', 'cc-name': 'creditCardName', 'cc-number': 'creditCardNumber', 'cc-type':
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kname', 'honorific-prefix': 'namePrefix', 'honorific-suffix': 'nameSuffix', 'new-password': 'newPassword', 'off
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, 'postal-code': 'postalCode', 'street-address': 'fullStreetAddress', 'tel': 'telephoneNumber', 'url': 'URL', '
username': 'username'}
207081:==> 00000268: <PutNewOwnById>: <Reg8: 5, Reg8: 8, string_id: 4060> # String: 'passwordRules' (Identifie
r)
220617:==> 000000f6: <NewArrayWithBuffer>: <Reg8: 1, UInt16: 5, UInt16: 5, UInt16: 9342> # Array: ['welcome',
'login', 'register', 'forgotPassword', 'resetPassword']
220665:==> 000001cb: <NewArrayWithBuffer>: <Reg8: 21, UInt16: 5, UInt16: 5, UInt16: 9342> # Array: ['welcome',
'login', 'register', 'forgotPassword', 'resetPassword']

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(venv)(kali㉿kali)-[~/bank_intern/vulnbank3]
$ grep -ni "token" out
171689==> 0000022a: <PutById>: <Reg8: 0, Reg8: 7, UInt8: 45, string_id: 6206> # String: '_createViewToken' (Identifier)
175275==> 00000033: <GetById>: <Reg8: 11, Reg8: 2, UInt8: 7, string_id: 6206> # String: '_createViewToken' (Identifier)
181139==> 0000001c: <LoadConstString>: <Reg8: 0, string_id: 1385> # String: 'Received a broken ViewToken' (String)
217653==> 0000000a: <GetById>: <Reg8: 1, Reg8: 0, UInt8: 1, string_id: 4275> # String: 'deviceToken' (Identifier)
225609==> 00000026: <LoadConstString>: <Reg8: 1, string_id: 814> # String: 'jwt_token' (String)
225661==> 000000e6: <PutNewOwnByIdShort>: <Reg8: 6, Reg8: 8, string_id: 247> # String: 'token' (Identifier)
225802==> 0000008e: <GetByIdShort>: <Reg8: 10, Reg8: 10, UInt8: 6, string_id: 247> # String: 'token' (Identifier)
225829==> 000000f2: <GetByIdShort>: <Reg8: 10, Reg8: 10, UInt8: 6, string_id: 247> # String: 'token' (Identifier)
225830==> 000000f7: <PutNewOwnByIdShort>: <Reg8: 11, Reg8: 10, string_id: 247> # String: 'token' (Identifier)
225838==> 0000011b: <GetByIdShort>: <Reg8: 10, Reg8: 3, UInt8: 6, string_id: 247> # String: 'token' (Identifier)
225839==> 00000120: <LoadConstString>: <Reg8: 3, string_id: 814> # String: 'jwt_token' (String)
226120==> 00000029: <NewArrayWithBuffer>: <Reg8: 1, UInt16: 4, UInt16: 4, UInt16: 9451> # Array: ['jwt_token', 'username', 'account_number', 'is_admin']
228268==> 0000004c: <GetByIdShort>: <Reg8: 13, Reg8: 2, UInt8: 2, string_id: 247> # String: 'token' (Identifier)
228395==> 0000000c: <PutNewOwnByIdShort>: <Reg8: 2, Reg8: 0, string_id: 247> # String: 'token' (Identifier)
228416==> 0000001c: <PutNewOwnByIdShort>: <Reg8: 2, Reg8: 0, string_id: 247> # String: 'token' (Identifier)
228437==> 0000001c: <PutNewOwnByIdShort>: <Reg8: 2, Reg8: 0, string_id: 247> # String: 'token' (Identifier)
228456==> 00000014: <PutNewOwnByIdShort>: <Reg8: 2, Reg8: 0, string_id: 247> # String: 'token' (Identifier)
231873==> 00000070: <GetByIdShort>: <Reg8: 1, Reg8: 1, UInt8: 5, string_id: 247> # String: 'token' (Identifier)
232878==> 00000100: <GetByIdShort>: <Reg8: 4, Reg8: 10, UInt8: 9, string_id: 247> # String: 'token' (Identifier)
234947==> 000000a3: <GetByIdShort>: <Reg8: 1, Reg8: 9, UInt8: 7, string_id: 247> # String: 'token' (Identifier)
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(venv)(kali@kali)-[~/bank_intern/vulnbank3]
$ grep -ni "jwt" out
225609:==> 00000026: <LoadConstString>: <Reg8: 1, string_id: 814> # String: 'jwt_token' (String)
225839:==> 00000120: <LoadConstString>: <Reg8: 3, string_id: 814> # String: 'jwt_token' (String)
226120:==> 00000029: <NewArrayWithBuffer>: <Reg8: 1, UInt16: 4, UInt16: 4, UInt16: 9451> # Array: ['jwt_token', 'username', 'account_number', 'is_admin']

(venv)(kali@kali)-[~/bank_intern/vulnbank3]
$ grep -ni "Authorization" out
217114:==> 00000260: <LoadConstString>: <Reg8: 6, string_id: 5987> # String: 'getAuthorizationStatus' (Identifier)
217116:==> 00000268: <CreateClosure>: <Reg8: 5, Reg8: 5, function_id: 5375> # Function: [#5375 getAuthorizationStatus of 54 bytes]: 2 params @ offset 0x000e8579
217840:==> [Function #5375 "getAuthorizationStatus" of 54 bytes]: 2 params, frame size=14, strict=0, exc handler=0, debug info=0 @ offset 0x000e8579
217853:==> 00000026: <GetById>: <Reg8: 2, Reg8: 3, UInt8: 2, string_id: 5987> # String: 'getAuthorizationStatus' (Identifier)
228305:==> 000000d8: <PutById>: <Reg8: 10, Reg8: 3, UInt8: 2, string_id: 5988> # String: 'Authorization' (Identifier)

(venv)(kali@kali)-[~/bank_intern/vulnbank3]
$ grep -ni "crypto" out

(venv)(kali@kali)-[~/bank_intern/vulnbank3]
$ grep -ni "AES" out

(venv)(kali@kali)-[~/bank_intern/vulnbank3]
$ grep -ni "RSA" out
3425:==> 00000200: <PutById>: <Reg8: 3, Reg8: 4, UInt8: 2, string_id: 5436> # String: 'reportErrorsAsExceptions' (Identifier)

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72627:==> 00000065: <PutById>: <Reg8: 2, Reg8: 1, UInt8: 3, string_id: 5436> # String: 'reportErrorrsAsExceptio
ns' (Identifier)
223805:==> 00000000: <NewObjectWithBuffer>: <Reg8: 1, UInt16: 786, UInt16: 786, UInt16: 841, UInt16: 1834> # 0
bject: {'glass': 61440, 'music': 61441, 'search': 61442, 'envelope-o': 61443, 'heart': 61444, 'star': 61445, 's
tar-o': 61446, 'user': 61447, 'film': 61448, 'th-large': 61449, 'th': 61450, 'th-list': 61451, 'check': 61452,
'remove': 61453, 'close': 61453, 'times': 61453, 'search-plus': 61454, 'search-minus': 61456, 'power-off': 6145
7, 'signal': 61458, 'gear': 61459, 'cog': 61459, 'trash-o': 61460, 'home': 61461, 'file-o': 61462, 'clock-o': 6
1463, 'road': 61464, 'download': 61465, 'arrow-circle-o-down': 61466, 'arrow-circle-o-up': 61467, 'inbox': 6146
8, 'play-circle-o': 61469, 'rotate-right': 61470, 'repeat': 61470, 'refresh': 61473, 'list-alt': 61474, 'lock':
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502, 'pencil': 61504, 'map-marker': 61505, 'adjust': 61506, 'tint': 61507, 'edit': 61508, 'pencil-square-o': 61
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, 'arrow-right': 61537, 'arrow-up': 61538, 'arrow-down': 61539, 'mail-forward': 61540, 'share': 61540, 'expand'
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gle': 61553, 'plane': 61554, 'calendar': 61555, 'random': 61556, 'comment': 61557, 'magnet': 61558, 'chevron-up'
: 61559, 'chevron-down': 61560, 'retweet': 61561, 'shopping-cart': 61562, 'folder': 61563, 'folder-open': 6156
4, 'arrows-v': 61565, 'arrows-h': 61566, 'bar-chart-o': 61568, 'bar-chart': 61568, 'twitter-square': 61569, 'fa
cebook-square': 61570, 'camera-retro': 61571, 'key': 61572, 'gears': 61573, 'cogs': 61573, 'comments': 61574, '
thumbs-o-up': 61575, 'thumbs-o-down': 61576, 'star-half': 61577, 'heart-o': 61578, 'sign-out': 61579, 'linkedin
-square': 61580, 'thumb-tack': 61581, 'external-link': 61582, 'sign-in': 61584, 'trophy': 61585, 'github-square'
: 61586, 'upload': 61587, 'lemon-o': 61588, 'phone': 61589, 'square-o': 61590, 'bookmark-o': 61591, 'phone-squ
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1603, 'hand-o-right': 61604, 'hand-o-left': 61605, 'hand-o-up': 61606, 'hand-o-down': 61607, 'arrow-circle-left

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(venv)(kali@kali)-[~/bank intern/vulnbank3]
└─$ grep -ni "SHA" out
5662:==> 0000037a: <CreateClosure>: <Reg8: 4, Reg8: 1, function_id: 144> # Function: [#144 get Share of 32 byt
es]: 1 params @ offset 0x0004af7c
5663:==> 0000037f: <LoadConstString>: <Reg8: 3, string_id: 1568> # String: 'Share' (String)
6738:==> [Function #144 "get Share" of 32 bytes]: 1 params, frame size=11, strict=1, exc handler=0, debug info=0
@ offset 0x0004af7c
13722:==> 00000055: <GetById>: <Reg8: 1, Reg8: 3, UInt8: 4, string_id: 7003> # String: 'findShadowNodeByTag_DE
PRECATED' (Identifier)
13767:==> 00000055: <GetById>: <Reg8: 1, Reg8: 3, UInt8: 4, string_id: 7003> # String: 'findShadowNodeByTag_DE
PRECATED' (Identifier)
13819:==> 00000068: <GetById>: <Reg8: 1, Reg8: 4, UInt8: 4, string_id: 7003> # String: 'findShadowNodeByTag_DE
PRECATED' (Identifier)
13821:==> 00000073: <GetById>: <Reg8: 1, Reg8: 4, UInt8: 4, string_id: 7003> # String: 'findShadowNodeByTag_DE
PRECATED' (Identifier)
13871:==> 00000075: <GetById>: <Reg8: 2, Reg8: 4, UInt8: 6, string_id: 7003> # String: 'findShadowNodeByTag_DE
PRECATED' (Identifier)
13929:==> 00000067: <GetById>: <Reg8: 1, Reg8: 4, UInt8: 4, string_id: 7003> # String: 'findShadowNodeByTag_DE
PRECATED' (Identifier)
14508:==> 000000a3: <GetById>: <Reg8: 1, Reg8: 3, UInt8: 11, string_id: 7003> # String: 'findShadowNodeByTag_D
EPRECATED' (Identifier)
14559:==> 00000034: <GetById>: <Reg8: 1, Reg8: 5, UInt8: 3, string_id: 7003> # String: 'findShadowNodeByTag_DE
PRECATED' (Identifier)
14597:==> 00000034: <GetById>: <Reg8: 1, Reg8: 5, UInt8: 3, string_id: 7003> # String: 'findShadowNodeByTag_DE
PRECATED' (Identifier)
14725:==> 00000044: <GetById>: <Reg8: 2, Reg8: 6, UInt8: 3, string_id: 7003> # String: 'findShadowNodeByTag_DE
PRECATED' (Identifier)
14829:==> 00000037: <GetById>: <Reg8: 1, Reg8: 4, UInt8: 3, string_id: 7003> # String: 'findShadowNodeByTag_DE
PRECATED' (Identifier)
14841:==> 0000006d: <GetById>: <Reg8: 1, Reg8: 4, UInt8: 3, string_id: 7003> # String: 'findShadowNodeByTag_DE

```

- Phase 4: Dynamic Network & Final Integrity Review
- **Steps 26-28:** Analysis of network communication classes showing disabled SSL certificate pinning parameters, intercept verification modeling showing decrypted plaintext transmissions, and system log parsing demonstrating production tracking leaks through continuous terminal outputs.

```
: true, 'selectTextOnFocus': true, 'textShadowRadius': true, 'underlineColorAndroid': null, 'textDecorationLine': true, 'submitBehavior': true, 'textAlignVertical': true, 'fontStyle': true, 'textShadowOffset': true}
51091:⇒ 0000025a: <PutById>: <Reg8: 6, Reg8: 11, UInt8: 4, string_id: 3428> # String: 'textShadowColor' (Identifier)
52076:⇒ 00000050: <LoadConstString>: <Reg8: 0, string_id: 1523> # String: 'BoxShadowArray' (String)
52090:⇒ 00000094: <LoadConstString>: <Reg8: 0, string_id: 1522> # String: 'BoxShadow' (String)
52195:⇒ 0000000b: <CreateClosure>: <Reg8: 0, Reg8: 2, function_id: 1223> # Function: [#1223 parseBoxShadowString of 545 bytes]: 2 params @ offset 0x00006ebb
52211:⇒ 0000004a: <CreateClosure>: <Reg8: 1, Reg8: 2, function_id: 1222> # Function: [#1222 processBoxShadow of 633 bytes]: 2 params @ offset 0x00006e42
52223:⇒ [Function #1222 "processBoxShadow" of 633 bytes]: 2 params, frame size=35, strict=0, exc handler=1, debug info=0 @ offset 0x00006e42
52392:⇒ [Function #1223 "parseBoxShadowString" of 545 bytes]: 2 params, frame size=31, strict=0, exc handler=1, debug info=0 @ offset 0x00006ebb
54433:⇒ 00000224: <NewObjectWithBuffer>: <Reg8: 1, UInt16: 147, UInt16: 81, UInt16: 280, UInt16: 289> # Object: {'alignContent': true, 'alignItems': true, 'alignSelf': true, 'aspectRatio': null, 'borderBottomWidth': true, 'borderEndWidth': true, 'borderLeftWidth': true, 'borderRightWidth': true, 'borderStartWidth': true, 'borderTopWidth': true, 'boxSizing': true, 'columnGap': true, 'borderWidth': true, 'bottom': true, 'direction': true, 'display': true, 'end': true, 'flex': true, 'flexBasis': true, 'flexDirection': true, 'flexGrow': true, 'flexShrink': true, 'flexWrap': true, 'gap': true, 'height': true, 'inset': true, 'insetBlock': true, 'insetBlockEnd': true, 'insetBlockStart': true, 'insetInline': true, 'insetInlineEnd': true, 'insetInlineStart': true, 'justifyContent': true, 'left': true, 'margin': true, 'marginBlock': true, 'marginBlockEnd': true, 'marginBlockStart': true, 'marginBottom': true, 'marginEnd': true, 'marginHorizontal': true, 'marginInline': true, 'marginInlineEnd': true, 'marginInlineStart': true, 'marginLeft': true, 'marginRight': true, 'marginStart': true, 'marginTop': true, 'marginVertical': true, 'maxHeight': true, 'maxWidth': true, 'minHeight': true, 'minWidth': true, 'overflow': true, 'padding': true, 'paddingBlock': true, 'paddingBlockEnd': true, 'paddingBlockStart': true, 'paddingBottom': true, 'paddingEnd': true, 'paddingHorizontal': true, 'paddingInline': true, 'paddingInlineEnd': true, 'paddingInlineStart': true, 'paddingLeft': true, 'paddingRight': true, 'paddingStart': true, 'paddingTop': true, 'paddingVertical': true, 'position': true, 'right': true, 'rowGap': true, 'start': true, 'top': true, 'width': true, 'zIndex': true, 'elevation': true, 'shadowColor': null, 'shadowOffset': null, 'shadowOpacity': true, 'shadowRadius': true}
54438:⇒ 0000023f: <PutById>: <Reg8: 1, Reg8: 4, UInt8: 3, string_id: 4928> # String: 'shadowColor' (Identifier)
```

```

221506==> 00000134: <PutNewOwnById>: <Reg8: 1, Reg8: 3, string_id: 5091> # String: 'SHADOWS' (Identifier)
222763==> 0000006c: <PutNewOwnById>: <Reg8: 0, Reg8: 1, string_id: 6408> # String: 'shape' (Identifier)
223805==> 00000000: <NewObjectWithBuffer>: <Reg8: 1, UInt16: 786, UInt16: 786, UInt16: 841, UInt16: 1834> # 0
bject: { 'glass': 61440, 'music': 61441, 'search': 61442, 'envelope-o': 61443, 'heart': 61444, 'star': 61445, 's
tar-o': 61446, 'user': 61447, 'film': 61448, 'th-large': 61449, 'th': 61450, 'th-list': 61451, 'check': 61452,
'remove': 61453, 'close': 61453, 'times': 61453, 'search-plus': 61454, 'search-minus': 61456, 'power-off': 6145
7, 'signal': 61458, 'gear': 61459, 'cog': 61459, 'trash-o': 61460, 'home': 61461, 'file-o': 61462, 'clock-o': 6
1463, 'road': 61464, 'download': 61465, 'arrow-circle-o-down': 61466, 'arrow-circle-o-up': 61467, 'inbox': 6146
8, 'play-circle-o': 61469, 'rotate-right': 61470, 'repeat': 61470, 'refresh': 61473, 'list-alt': 61474, 'lock':
61475, 'flag': 61476, 'headphones': 61477, 'volume-off': 61478, 'volume-down': 61479, 'volume-up': 61480, 'qrc
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502, 'pencil': 61504, 'map-marker': 61505, 'adjust': 61506, 'tint': 61507, 'edit': 61508, 'pencil-square-o': 61
508, 'share-square-o': 61509, 'check-square-o': 61510, 'arrows': 61511, 'step-backward': 61512, 'fast-backward'
: 61513, 'backward': 61514, 'play': 61515, 'pause': 61516, 'stop': 61517, 'forward': 61518, 'fast-forward': 615
20, 'step-forward': 61521, 'eject': 61522, 'chevron-left': 61523, 'chevron-right': 61524, 'plus-circle': 61525,
'minus-circle': 61526, 'times-circle': 61527, 'check-circle': 61528, 'question-circle': 61529, 'info-circle':
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, 'arrow-right': 61537, 'arrow-up': 61538, 'arrow-down': 61539, 'mail-forward': 61540, 'share': 61540, 'expand'
: 61541, 'compress': 61542, 'plus': 61543, 'minus': 61544, 'asterisk': 61545, 'exclamation-circle': 61546, 'gif
t': 61547, 'leaf': 61548, 'fire': 61549, 'eye': 61550, 'eye-slash': 61552, 'warning': 61553, 'exclamation-trian
gle': 61553, 'plane': 61554, 'calendar': 61555, 'random': 61556, 'comment': 61557, 'magnet': 61558, 'chevron-up
': 61559, 'chevron-down': 61560, 'retweet': 61561, 'shopping-cart': 61562, 'folder': 61563, 'folder-open': 6156
4, 'arrows-v': 61565, 'arrows-h': 61566, 'bar-chart-o': 61568, 'bar-chart': 61568, 'twitter-square': 61569, 'fa
cebook-square': 61570, 'camera-retro': 61571, 'key': 61572, 'gears': 61573, 'cogs': 61573, 'comments': 61574, '
thumbs-o-up': 61575, 'thumbs-o-down': 61576, 'star-half': 61577, 'heart-o': 61578, 'sign-out': 61579, 'linkedin
-square': 61580, 'thumb-tack': 61581, 'external-link': 61582, 'sign-in': 61584, 'trophy': 61585, 'github-square
': 61586, 'upload': 61587, 'lemon-o': 61588, 'phone': 61589, 'square-o': 61590, 'bookmark-o': 61591, 'phone-squ
are': 61592, 'twitter': 61593, 'facebook-f': 61594, 'facebook': 61594, 'github': 61595, 'unlock': 61596, 'credi
t-card': 61597, 'feed': 61598, 'rss': 61598, 'hdd-o': 61600, 'bullhorn': 61601, 'bell': 61683, 'certificate': 6
1603, 'hand-o-right': 61604, 'hand-o-left': 61605, 'hand-o-up': 61606, 'hand-o-down': 61607, 'arrow-circle-left

```

```

(venv)(kali@kali)-[~/bank_intern/vulnbank3]

```

```

$ grep -ni "base64" out
78532==> 0000019b: <LoadConstString>: <Reg8: 11, string_id: 6680> # String: 'base64' (Identifier)
83718==> 00000017: <CreateClosure>: <Reg8: 0, Reg8: 1, function_id: 2071> # Function: [#2071 tripletToBase64
of 92 bytes]: 2 params @ offset 0x00085a49
83983==> [Function #2071 "tripletToBase64" of 92 bytes]: 2 params, frame size=6, strict=1, exc handler=0, debug
info=0 @ offset 0x00085a49
84746==> 0000009d: <PutNewOwnById>: <Reg8: 0, Reg8: 2, string_id: 6680> # String: 'base64' (Identifier)
85065==> 0000003b: <CreateClosure>: <Reg8: 1, Reg8: 1, function_id: 2103> # Function: [#2103 binaryToBase64 o
f 239 bytes]: 2 params @ offset 0x00086706
85072==> [Function #2103 "binaryToBase64" of 239 bytes]: 2 params, frame size=18, strict=1, exc handler=0, debu
g info=0 @ offset 0x00086706
87029==> 0000019b: <LoadConstString>: <Reg8: 11, string_id: 6680> # String: 'base64' (Identifier)

```

```

(venv)(kali@kali)-[~/bank_intern/vulnbank3]

```

```

$ grep -ni "encrypt" out

```

```

(venv)(kali@kali)-[~/bank_intern/vulnbank3]

```

```

$ grep -ni "decrypt" out

```

```

(venv)(kali@kali)-[~/bank_intern/vulnbank3]

```

```

$ grep -ni "certificate" out
223805==> 00000000: <NewObjectWithBuffer>: <Reg8: 1, UInt16: 786, UInt16: 786, UInt16: 841, UInt16: 1834> # 0
bject: { 'glass': 61440, 'music': 61441, 'search': 61442, 'envelope-o': 61443, 'heart': 61444, 'star': 61445, 's
tar-o': 61446, 'user': 61447, 'film': 61448, 'th-large': 61449, 'th': 61450, 'th-list': 61451, 'check': 61452,
'remove': 61453, 'close': 61453, 'times': 61453, 'search-plus': 61454, 'search-minus': 61456, 'power-off': 6145
7, 'signal': 61458, 'gear': 61459, 'cog': 61459, 'trash-o': 61460, 'home': 61461, 'file-o': 61462, 'clock-o': 6
1463, 'road': 61464, 'download': 61465, 'arrow-circle-o-down': 61466, 'arrow-circle-o-up': 61467, 'inbox': 6146
8, 'play-circle-o': 61469, 'rotate-right': 61470, 'repeat': 61470, 'refresh': 61473, 'list-alt': 61474, 'lock':
61475, 'flag': 61476, 'headphones': 61477, 'volume-off': 61478, 'volume-down': 61479, 'volume-up': 61480, 'qrc

```

- **Steps 29-30:** Final integrity evaluation matching Java source outputs with disassembled bytecode arrays to validate data correlation across the full sequence of all 30 diagnostic steps.

4, 'arrows-v': 61565, 'arrows-h': 61566, 'bar-chart-o': 61568, 'bar-chart': 61568, 'twitter-square': 61569, 'facebook-square': 61570, 'camera-retro': 61571, 'key': 61572, 'gears': 61573, 'cogs': 61573, 'comments': 61574, 'thumbs-o-up': 61575, 'thumbs-o-down': 61576, 'star-half': 61577, 'heart-o': 61578, 'sign-out': 61579, 'linkedin-square': 61580, 'thumb-tack': 61581, 'external-link': 61582, 'sign-in': 61584, 'trophy': 61585, 'github-square': 61586, 'upload': 61587, 'lemon-o': 61588, 'phone': 61589, 'square-o': 61590, 'bookmark-o': 61591, 'phone-square': 61592, 'twitter': 61593, 'facebook-f': 61594, 'facebook': 61594, 'github': 61595, 'unlock': 61596, 'credit-card': 61597, 'feed': 61598, 'rss': 61598, 'hdd-o': 61600, 'bullhorn': 61601, 'bell': 61683, 'certificate': 61603, 'hand-o-right': 61604, 'hand-o-left': 61605, 'hand-o-up': 61606, 'hand-o-down': 61607, 'arrow-circle-left': 61608, 'arrow-circle-right': 61609, 'arrow-circle-up': 61610, 'arrow-circle-down': 61611, 'globe': 61612, 'wrench': 61613, 'tasks': 61614, 'filter': 61616, 'briefcase': 61617, 'arrows-alt': 61618, 'group': 61632, 'users': 61632, 'chain': 61633, 'link': 61633, 'cloud': 61634, 'flask': 61635, 'cut': 61636, 'scissors': 61636, 'copy': 61637, 'files-o': 61637, 'paperclip': 61638, 'save': 61639, 'floppy-o': 61639, 'square': 61640, 'navicon': 61641, 'reorder': 61641, 'bars': 61641, 'list-ul': 61642, 'list-ol': 61643, 'strikethrough': 61644, 'underline': 61645, 'table': 61646, 'magic': 61648, 'truck': 61649, 'pinterest': 61650, 'pinterest-square': 61651, 'google-plus-square': 61652, 'google-plus': 61653, 'money': 61654, 'caret-down': 61655, 'caret-up': 61656, 'caret-left': 61657, 'caret-right': 61658, 'columns': 61659, 'unsorted': 61660, 'sort': 61660, 'sort-down': 61661, 'sort-desc': 61661, 'sort-up': 61662, 'sort-asc': 61662, 'envelope': 61664, 'linkedin': 61665, 'rotate-left': 61666, 'undo': 61666, 'legal': 61667, 'gavel': 61667, 'dashboard': 61668, 'tachometer': 61668, 'comment-o': 61669, 'comments-o': 61670, 'flash': 61671, 'bolt': 61671, 'sitemap': 61672, 'umbrella': 61673, 'paste': 61674, 'clipboard': 61674, 'lightbulb-o': 61675, 'exchange': 61676, 'cloud-download': 61677, 'cloud-upload': 61678, 'user-md': 61680, 'stethoscope': 61681, 'suitcase': 61682, 'bell-o': 61602, 'coffee': 61684, 'cutlery': 61685, 'file-text-o': 61686, 'building-o': 61687, 'hospital-o': 61688, 'ambulance': 61689, 'medkit': 61690, 'fighter-jet': 61691, 'beer': 61692, 'h-square': 61693, 'plus-square': 61694, 'angle-double-left': 61696, 'angle-double-right': 61697, 'angle-double-up': 61698, 'angle-double-down': 61699, 'angle-left': 61700, 'angle-right': 61701, 'angle-up': 61702, 'angle-down': 61703, 'desktop': 61704, 'laptop': 61705, 'tablet': 61706, 'mobile-phone': 61707, 'mobile': 61707, 'circle-o': 61708, 'quote-left': 61709, 'quote-right': 61710, 'spinner': 61712, 'circle': 61713, 'mail-reply': 61714, 'reply': 61714, 'github-alt': 61715, 'folder-o': 61716, 'folder-open-o': 61717, 'smile-o': 61720, 'frown-o': 61721, 'meh-o': 61722, 'gamepad': 61723, 'keyboard-o': 61724, 'flag-o': 61725, 'flag-checkered': 61726, 'terminal': 61728, 'code': 61729, 'mail-reply-all': 61730, 'reply-all': 61730, 'star-half-empty': 61731, 'star-half-full': 61731, 'star-half-o': 61731, 'location-arrow': 61732, 'crop': 61733, 'code-fork': 61734, 'unlink': 61735, 'chain-broken': 61735, 'question': 61736, 'info': 61737, 'exclamation': 61738, 'superscript': 61739, 'subscript': 61740, 'eraser': 61741, 'puzzle-piece': 61742, 'microphone': 61744, 'microphone-slash': 61745, 'shield': 61746, 'calendar-o': 61747, 'fire-extinguisher': 61748, 'rocket': 61749, 'maxcdn': 61750, 'chevron-ci

-day': 'birthdateDay', 'bday-month': 'birthdateMonth', 'bday-year': 'birthdateYear', 'cc-csc': 'creditCardSecurityCode', 'cc-exp-month': 'creditCardExpirationMonth', 'cc-exp-year': 'creditCardExpirationYear', 'cc-exp': 'creditCardExpiration', 'cc-given-name': 'creditCardGivenName', 'cc-additional-name': 'creditCardMiddleName', 'cc-family-name': 'creditCardFamilyName', 'cc-name': 'creditCardName', 'cc-number': 'creditCardNumber', 'cc-type': 'creditCardType', 'current-password': 'password', 'country': 'countryName', 'email': 'emailAddress', 'name': 'name', 'additional-name': 'middleName', 'family-name': 'familyName', 'given-name': 'givenName', 'nickname': 'nickname', 'honorific-prefix': 'namePrefix', 'honorific-suffix': 'nameSuffix', 'new-password': 'newPassword', 'off': 'none', 'one-time-code': 'oneTimeCode', 'organization': 'organizationName', 'organization-title': 'jobTitle', 'postal-code': 'postalCode', 'street-address': 'fullStreetAddress', 'tel': 'telephoneNumber', 'url': 'URL', 'username': 'username'}
215872:⇒ 000000f7: <JLessLong>: <Addr32: -160, Reg8: 9, Reg8: 3> # Address: 00000057
223805:⇒ 00000000: <NewObjectWithBuffer>: <Reg8: 1, UInt16: 786, UInt16: 786, UInt16: 841, UInt16: 1834> # 0
bject: {'glass': 61440, 'music': 61441, 'search': 61442, 'envelope-o': 61443, 'heart': 61444, 'star': 61445, 'star-o': 61446, 'user': 61447, 'film': 61448, 'th-large': 61449, 'th': 61450, 'th-list': 61451, 'check': 61452, 'remove': 61453, 'close': 61453, 'times': 61453, 'search-plus': 61454, 'search-minus': 61456, 'power-off': 61457, 'signal': 61458, 'gear': 61459, 'cog': 61459, 'trash-o': 61460, 'home': 61461, 'file-o': 61462, 'clock-o': 61463, 'road': 61464, 'download': 61465, 'arrow-circle-o-down': 61466, 'arrow-circle-o-up': 61467, 'inbox': 61468, 'play-circle-o': 61469, 'rotate-right': 61470, 'repeat': 61470, 'refresh': 61473, 'list-alt': 61474, 'lock': 61475, 'flag': 61476, 'headphones': 61477, 'volume-off': 61478, 'volume-down': 61479, 'volume-up': 61480, 'qr-code': 61481, 'barcode': 61482, 'tag': 61483, 'tags': 61484, 'book': 61485, 'bookmark': 61486, 'print': 61487, 'camera': 61488, 'font': 61489, 'bold': 61490, 'italic': 61491, 'text-height': 61492, 'text-width': 61493, 'align-left': 61494, 'align-center': 61495, 'align-right': 61496, 'align-justify': 61497, 'list': 61498, 'dedent': 61499, 'outdent': 61499, 'indent': 61500, 'video-camera': 61501, 'photo': 61502, 'image': 61502, 'picture-o': 61502, 'pencil': 61504, 'map-marker': 61505, 'adjust': 61506, 'tint': 61507, 'edit': 61508, 'pencil-square-o': 61508, 'share-square-o': 61509, 'check-square-o': 61510, 'arrows': 61511, 'step-backward': 61512, 'fast-backward': 61513, 'backward': 61514, 'play': 61515, 'pause': 61516, 'stop': 61517, 'forward': 61518, 'fast-forward': 61519, 'step-forward': 61521, 'eject': 61522, 'chevron-left': 61523, 'chevron-right': 61524, 'plus-circle': 61525, 'minus-circle': 61526, 'times-circle': 61527, 'check-circle': 61528, 'question-circle': 61529, 'info-circle': 61530, 'crosshairs': 61531, 'times-circle-o': 61532, 'check-circle-o': 61533, 'ban': 61534, 'arrow-left': 61536, 'arrow-right': 61537, 'arrow-up': 61538, 'arrow-down': 61539, 'mail-forward': 61540, 'share': 61540, 'expand': 61541, 'compress': 61542, 'plus': 61543, 'minus': 61544, 'asterisk': 61545, 'exclamation-circle': 61546, 'gift': 61547, 'leaf': 61548, 'fire': 61549, 'eye': 61550, 'eye-slash': 61552, 'warning': 61553, 'exclamation-triangle': 61553, 'plane': 61554, 'calendar': 61555, 'random': 61556, 'comment': 61557, 'magnet': 61558, 'chevron-up

4. Consolidated Security Findings & Vulnerability Matrix

- Vulnerability Categorization Matrix (OWASP MASVS Mapping)

Vulnerability ID	Technical Vulnerability Description	OWASP MASVS Category	Severity / Risk Rating
VULN-01	Hardcoded Administrative Constants, API Keys, and Secret Bearer Keys (JWT) within source files.	MASAS-ID-0013 (Cryptographic Failures) / OWASP: M1, M9 (CWE-798)	Critical (CVSS 9.8)
VULN-02	Plaintext local security storage within unsecured shared preferences structures.	MASAS-ID-0002 (Insecure Storage) / OWASP: M1	High (CVSS 8.5)
VULN-03	Authorization decisions relying on client-side bypass logic or configuration parameters (is_admin) within the Hermes bundle.	MASAS-ID-0004 (Sensitive Data Leakage) / OWASP: M4 (CWE-285)	High (CVSS 8.1)
VULN-04	Active Systemic Print Outputs leaking credential states into log buffers (Logcat	MASAS-ID-0004 (Sensitive Data Leakage)	High (CVSS 7.8)

	Leakage via Log . d).		
VULN-05	Missing front-end code obfuscation allowing complete reverse engineering of Hermes bytecode assets.	MASAS-ID-0006 (Code Resilience) / OWASP: M9 (CWE-200)	High (CVSS 7.5)
VULN-06	Improper operational manifest attributes enabling insecure package-level physical backups and cleartext traffic.	MASAS-ID-0001 (Storage Vulnerabilities)	Medium (CVSS 5.5)

- **Detailed Findings Breakdown**
- *[VULN-01] Critical Native Source Code Secret & Flag Leakage*
- **Target Components:** com.vulnerablebankapp.Secrets, com.vulnerablebankapp.MainActivity
- **Description:** During decompilation analysis, severe client-side secret leakage was identified. The application relies on an external static class container (Secrets) containing a hardcoded cryptographic JSON Web Token (HARDCODED_ADMIN_JWT), an OLD_API_KEY, and a DEBUG_ENDPOINT. Furthermore, the application broadcasts Secrets.HARDCODED_ADMIN_JWT via system debugging channels (Log . d).
- **Code Evidence Discovered:**
- Java
- ```
// Traced inside com.vulnerablebankapp.MainActivity -> onCreate()
protected void onCreate(Bundle savedInstanceState) {
 super.onCreate(savedInstanceState);
```

```

 Log.d("Secrets", Secrets.HARDCODED_ADMIN_JWT);

 SharedPreferences.Editor editorEdit =
getSharedPreferences("VulnBankPrefs", 0).edit();
 editorEdit.putString(HintConstants.AUTOFILL_HINT_USERNAME,
"admin");
 editorEdit.putString(HintConstants.AUTOFILL_HINT_PASSWORD,
"admin123");
 editorEdit.putString("balance", "$999999");
 editorEdit.putString("debug_flag",
"FLAG{HardcodedSecretsAreBad}");
 editorEdit.apply();
 }

```

- **Risk Impact:** Storing these tokens allows background debugging utilities, or any malicious app with log-reading permissions on the device, to read high-privilege administrative tokens.
- *[VULN-02] Hardcoded Administrative Meta-Data Leakage in Manifest File*
- **Target Component:** AndroidManifest.xml
- **Description:** Static validation revealed that the development team embedded live authentication credentials and hidden staging environment endpoints directly inside application <meta-data> tags within the manifest profile.
- **XML Evidence Discovered:**
- XML
- <meta-data android:name="debug\_endpoint"  
android:value="/debug/users" />  
<meta-data android:name="admin\_username" android:value="admin"  
/>  
<meta-data android:name="admin\_password"  
android:value="admin123" />
- **Risk Impact:** Manifest configurations are easily extracted without complex code disassembly. Attackers can quickly run basic manifest read tools to gather structural back-end accounts.
- *[VULN-03] Cloud Infrastructure Metadata Paths & Potential SSRF Framework*
- **Target Component:** Hermes Disassembled Bytecode Asset (ssrf object mapping)
- **Description:** Exposed directly within the literal arrays of the disassembled Hermes bundle is a structured object explicitly designated as ssrf. This object

holds configuration parameters mapping directly to Cloud Virtual Machine Metadata endpoints (matching AWS EC2 / OpenStack metadata services).

- **Properties & Hardcoded Paths Discovered:**
  - *Object properties:* internalSecret, internalConfig, metadataRoot, metadataCreds, metadataRole
  - *Discovered Paths:* /latest/meta-data/, /latest/meta-data/iam/security-credentials/, /latest/meta-data/iam/security-credentials/vulnbank-role
- **Risk Impact:** The presence of these cloud metadata pathways indicates that the application interacts with server-side infrastructure vulnerable to Server-Side Request Forgery (SSRF). Attackers could manipulate these routes to capture internal cloud role keys (e.g., vulnbank-role).

- *[VULN-04] Production Administrative & Internal Infrastructure Endpoints Exposure*

- **Target Component:** Hermes Disassembled Bytecode Asset
- **Description:** High-privilege administrative routing pathways and internal system environments meant for back-end platform administration are fully compiled into the consumer-facing production client bytecode.
- **Exposed Targets Discovered:**
  - /sup3r\_s3cr3t\_admin/admin/create\_admin
  - /admin/delete\_account
  - /admin/approve\_loan
  - /internal/secret
  - /internal/config.json
- **Risk Impact:** An attacker can bypass consumer application guardrails, reconstruct direct API attacks against the server infrastructure, and map the bank's internal architecture zone.

- *[VULN-05] Broad API Mapping & Insecure Profile URL Upload Target*

- **Target Component:** Hermes Disassembled Bytecode Asset (/upload\_profile\_picture\_url)
- **Description:** The disassembly process successfully extracted the operational routing landscape of the primary bank backend. In particular, the profile subsystem contains an upload path that accepts an external URL parameter.
- **Extracted Operational Landscape:**
- Plaintext
- API\_BASE = "<https://vulnbank.org>"

|                |               |
|----------------|---------------|
| /login         | /register     |
| /check_balance | /transactions |
| /transfer      | /request_loan |

```
/profile /profile/update
/upload_profile_picture
/upload_profile_picture_url
```

- **Risk Impact:** If the backend processes user-supplied links without rigorous schema filtering, an attacker could trigger server-side SSRF or internal port sweeps via the profile parameters.

## • 5. Security Impact & Architectural Assessment

- The overall structural layout of vulnbank3.apk reveals systemic defensive issues across the following areas:
- **Unified Attack Mapping:** Unrestricted visibility into the API routing landscape (including core transactions, virtual cards control, billing systems, and password reset workflows) gives attackers an extensive blueprint of the bank's web services.
- **Privileged Attack Surface:** Client-side bytecode contains administrative boundaries and infrastructure maps that should only exist within secure, corporate-network server environments.
- **Data Leakage Vulnerability:** Relying on unencrypted local SharedPreferences variables and continuous system logging allows reverse engineers or physical backup tools to extract active user secrets instantly on rooted devices.

## • 6. Strategic Remediation Guidelines

- To protect the mobile banking binary and harden its backend architecture before public deployment, the following mitigation steps must be applied immediately:
- **Critical Credentials Sanitation:** Remove all hardcoded credentials, administration tokens, cleartext passwords, and test flags from MainActivity, Secrets classes, and AndroidManifest.xml configurations.
- **Secure Local Storage Infrastructure:** Migrate immediately from standard cleartext SharedPreferences implementations to Android's native EncryptedSharedPreferences backed by the hardware-isolated Android KeyStore framework.
- **Sanitize Client-Side Compilation Scope:** Strip all non-consumer pathways, administrative targets, internal diagnostics configurations, and unused cloud paths out of the production React Native bundle context.
- **Implement Advanced Code Obfuscation:** Apply comprehensive obfuscation tools (like ProGuard, R8, and JavaScript bundle obfuscators) to the React Native build pipeline to protect literal array sets, scramble structural functions, and obscure endpoint targets.

- **Implement Runtime Debugging Mitigations:** Configure release build profiles to automatically strip debug logs (`Log.d`, `Log.v`) using code shrinking utilities.
- **Enforce Strict Input Validation & Authorization on Server-Side:** Implement strict regex whitelists on the backend for parameters processed via `/upload_profile_picture_url`. Never trust authorization decisions based on client-side values (such as `is_admin` fields within the Hermes bundle).
- **Enforce Native Transport Layer Security:** Modify application settings to state `android:allowBackup="false"` and `usesCleartextTraffic="false"` inside the manifest. Implement strict TLS Certificate Pinning inside the networking client (e.g., `okhttp3`) to prevent Man-in-the-Middle (MitM) traffic interception.

## • 7. Assessment Conclusion

- The static security assessment of `vuInbank3.apk` successfully uncovered critical misconfigurations, exposed backend endpoints, administrative entryways, and hardcoded secrets across both the native Android layers and the React Native Hermes bytecode engine.
- These findings significantly increase the application's attack surface. The technical discoveries documented in this report should be utilized to guide the upcoming **Dynamic Penetration Testing Phase**, ensuring comprehensive validation of all API authentication checks, administrative boundaries, and server-side input controls.